Monte Carlo Control vs Sarsa(λ) on Text Flappy Bird

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Abstract. This report compares two tabular reinforcement learning agents on the simple Text Flappy Bird environment: first-visit Monte Carlo control and Sarsa(λ). I use the distance-based state because it is small enough for a value-table approach. In my experiments, both agents learn, Monte Carlo improves faster early in training, and Sarsa(λ) reaches the best final score with a longer training budget.

Keywords: reinforcement learning · Monte Carlo control · Sarsa(λ)

1 Setup

I used the simple `TextFlappyBird-v0` environment from the course repository [1]. The state is (dx, dy), where dx is the horizontal distance to the next pipe gap and dy is the vertical distance between the bird and the center of that gap. The action space has two actions: idle and flap.

All main experiments use the default configuration `height=15, width=20, pipe_gap=4`. Both agents were trained for 2000 episodes, with a maximum of 200 steps per episode, and greedy evaluation every 200 episodes on 30 episodes. I measured return and score.

The Monte Carlo agent used $\gamma = 0.99$, $\epsilon = 0.20$, ϵ decay $= 0.999$, and minimum $\epsilon = 0.01$. The Sarsa(λ) agent used $\alpha = 0.15$, $\lambda = 0.9$, $\gamma = 0.99$, $\epsilon = 0.10$, ϵ decay $= 0.999$, and minimum $\epsilon = 0.02$.

The submitted notebook is available at GitHub notebook link.

2 Results

Table 1 shows the final evaluation after 2000 episodes. Both agents learned useful policies. Sarsa(λ) reached the best final result, with mean return 175.07 and mean score 16.50. Monte Carlo was close behind, with mean return 168.70 and score 16.00.

The learning curves in Fig. 1 show the main difference. Monte Carlo starts improving earlier and stays ahead during the first part of training. Sarsa(λ) improves more slowly at first, but by the end of training it slightly overtakes Monte Carlo in both return and score.

I also looked at the learned action-values and a small parameter sweep. Figure 2 shows the learned $Q(s, \text{flap})$ maps for both agents together with the sweep

Table 1. Final greedy evaluation after 2000 episodes.

Agent	Mean Return	Mean Score
Monte Carlo control	168.70	16.00
Sarsa(λ)	175.07	16.50

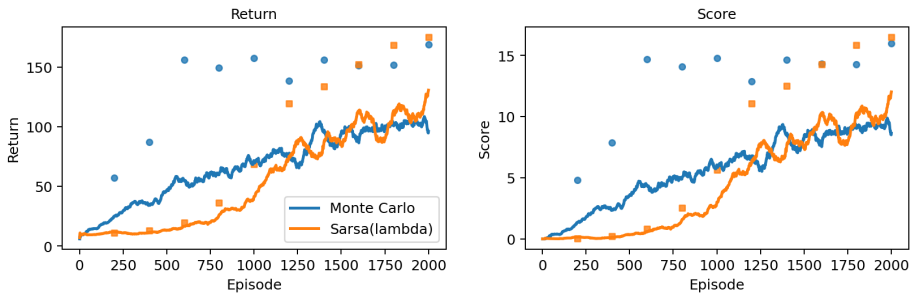

Fig. 1. Comparison of the training curves for both agents. The lines are 100-episode moving averages and the markers are greedy evaluations.

results. In both agents, the high-value states form a band around the safe region near the next gap.

3 Discussion

The sweep results were useful. For Monte Carlo, higher initial exploration helped: the best tested setting was $\epsilon = 0.20$, with mean return 175.40 and score 16.55. For Sarsa(λ), the parameter choice mattered more. A weak setting with $\alpha = 0.05$ and $\lambda = 0.7$ almost did not learn, while $\alpha = 0.15$ and $\lambda = 0.9$ worked much better and reached mean return 130.8 with score 12.0.

The two versions of the Text Flappy Bird environment are useful for different reasons. The simple version is suitable for tabular RL because the state is small. The screen version keeps more information, but the state space is much larger, so it would be a better fit for function approximation or deep RL. The same issue appears in the original Flappy Bird gym [3]: the RGB version is too large for a value table, and even the simple original environment would need discretization because its observations are continuous.

I also tested transfer to a few modified level settings. Monte Carlo scored 14.77 on the training layout, then dropped to 9.67 on a taller map and 5.93 on a wider map, but increased to 17.13 when the pipe gap was larger. Sarsa showed similar behavior, with 17.73 on the base layout, 9.50 on the taller map, 9.57 on the wider map, and 17.50 on the larger-gap setting. So both agents generalize partly, but both are still adapted to the training geometry.

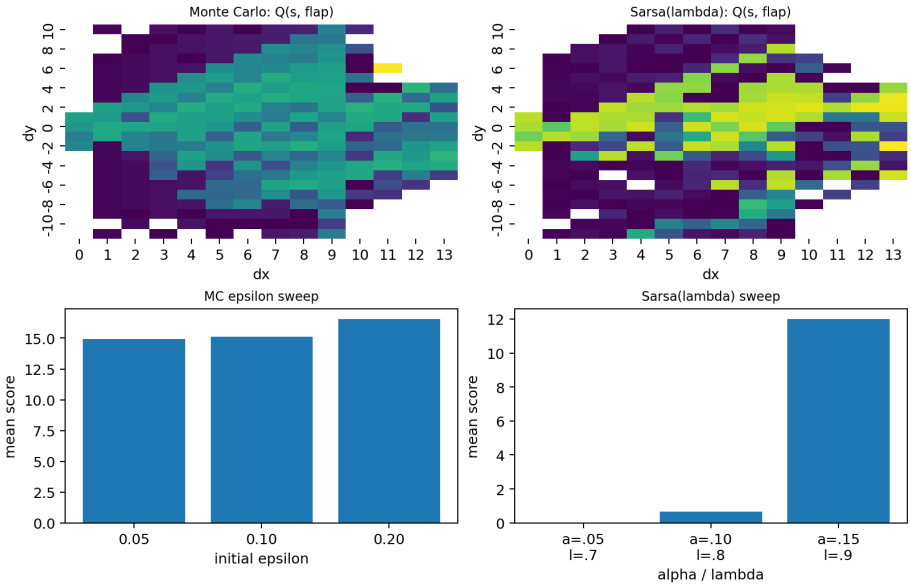

Fig. 2. Action-value summaries and compact parameter sweeps. Top: learned $Q(s, \text{flap})$ maps. Bottom: score from the tested sweep settings.

4 Conclusion

Overall, both methods learned a reasonable policy on the simple Text Flappy Bird environment. In my experiments, Monte Carlo was stronger early in training, but Sarsa(λ) became slightly better with the longer 2000 episode budget. Sarsa(λ) was still more sensitive to tuning. The simple (dx, dy) environment works for tabular RL, while richer versions of the game would likely need different methods.

References

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2. Sutton, R.S., Barto, A.G.: Reinforcement Learning: An Introduction. MIT Press, 2nd edn. (2018)
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